

Three-File Translation & Glossary Generator Pipeline

Overview

The **Three-File Translation & Glossary Generator** is an automated Python and Google Colab pipeline designed for translators, interpreters, language learners, and researchers. It takes any raw text input, executes true neural machine translation (supporting bidirectional English-Ukrainian and automatic language detection), extracts up to 100 key terminology items using advanced natural language processing (spaCy), and compiles everything into **three professional, separate Microsoft Word documents (.docx)** bundled into a single downloadable .zip archive.

Key Features

1. **True Neural Machine Translation:** Utilizes deep-translator with Google Translate backing to accurately convert multi-paragraph texts rather than echoing identical source text.
2. **Automated Language Handling:** Seamlessly switches between English-to-Ukrainian and Ukrainian-to-English translation pipelines.
3. **Smart Linguistic Glossary Extraction:** Powered by spaCy (en_core_web_sm / uk_core_news_sm), it extracts high-value noun chunks, proper nouns, and technical adjectives, filtering out common stop words and frequency-sorting the top terms.
4. **Three Distinct Output Artifacts:**
 - **1_Original_Document.docx:** Clean copy of the source text paragraphs.
 - **2_Translated_Document.docx:** Paragraph-by-paragraph translated output.
 - **3_Glossary_Terms.docx:** A structured table pairing source terms with their target translations.
5. **Instant ZIP Packaging:** Automatically compresses all three documents into translation_package.zip and triggers an instant browser download via Google Colab utilities.

Technical Stack & Dependencies

- **Python 3.13+**
- **python-docx:** Programmatic generation and formatting of Microsoft Word documents and grid tables.
- **spacy:** Industrial-strength Natural Language Processing for tokenization, noun-chunk extraction, and POS tagging.
- **deep-translator:** Robust wrapper for free translation APIs with auto-chunking for long text blocks.
- **google.colab:** Seamless file management and browser download triggering inside Google

Colab notebooks.

Pipeline Workflow

1. **Initialization & Setup:** Installs required packages (python-docx, spacy, deep-translator) and downloads appropriate language models (en_core_web_sm, uk_core_news_sm).
2. **Text Processing & Translation:** Splits input text into discrete paragraphs and translates them through the neural translator, handling character chunking safely.
3. **Glossary Extraction:** Runs spaCy over the source text, gathers candidate noun phrases and key terms, counts frequencies, and translates each entry individually.
4. **Document Generation:** Constructs the three Word documents with custom headings, paragraphs, and formatted grid tables (Table Grid).
5. **Archiving & Delivery:** Compresses the files into translation_package.zip and initiates automatic download.